Intelligent University Portal with QR Attendance and Predictive Analytics

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***Abstract*- The traditional university attendance system is still highly manual and requires 5 to 10 minutes of class time per session to perform administrative functions without creating any meaningful academic insight. While the current attendance systems either perform QR-based attendance or perform prediction separately, no system combines both functions and offers real-time classroom monitoring, and academic recommendation based on the collected data. This research paper introduces the Intelligent University Portal which brings together such a combination in one unified platform including QR-based attendance with limited period and manual code entry, attendance percentage, and predictive analytics to detect at-risk students based on attendance percentage. This portal caters for the use by admins, teachers, and students via respective dashboards with respective academic features with conversational chatbot.**

***Keywords*- Academic Monitoring, Academic Performance System, Attendance Analytics, Predictive Analytics, QR Attendance**

I. INTRODUCTION

The conventional way of monitoring attendance in higher education institutions involves conducting manual roll calls and taking sign-in forms, which takes about 5-10 minutes of teaching time for each class and generates paperwork but no information for analysis [1], [18]. Even though the usage of QR codes in attendance collection has been developed in order to enhance efficiency, the generated attendance records still do not translate into any academic insight [2], [3]. Simultaneously, the predictive analytics platform works in isolation, using only the student's performance metrics and ignoring the real-time attendance metrics that could be the early signs of their academic deterioration [7], [8]. The learning analytics dashboards offer historical data rather than real-time intervention options [15], [16].

The proposed Intelligent University Portal addresses this problem through a comprehensive web-based platform which integrates three key features such as secure QR attendance system with time-expiring codes, manual codes for those students who do not have cameras and compliance with Myanmar university attendance percentage formula. The system supports admins, teachers and students providing them with university-wide analytics for admins, classroom control and interventions for teachers and personalized dashboards and recommendations for students.

II. LITERATURE REVIEW

This section reviews recent studies related to QR code-based attendance systems, predictive learning analytics, and learning analytics dashboards in higher education. Existing research often focuses on individual aspects rather than providing an integrated solution for secure academic evaluation, predictive risk assessment, and real-time monitoring. The following review summarizes the current state of research and identifies the research gap addressed by the proposed system.

A. *QR-Code Based Attendance Management Systems*

QR Code technology is a cheap and practical way of handling attendance using the students' mobile phones [1]. Modern literature has addressed security issues through the use of dynamic QR codes and encryption, alongside the implementation of methods like NFC, facial recognition, and geolocation [2], [5]. Although the use of these technologies has improved the management of attendance, the vast majority still addresses only the aspects of attendance capturing and authentication [18], [19].

B. *Predictive Learning Analytics for Student Risk Assessment*

Predictive Learning Analytics detects students who are likely to have academic challenges through the analysis of educational data. Modern research employs machine learning algorithms including Random Forest, XGBoost, Support Vector Machine, deep learning, and Explainable Artificial Intelligence to predict academic achievement and retention [7], [14]. While these methods enhance student services, their application is mostly limited to academic data from past records and almost never involves real-time attendance data in an integrated academic management system.

C. *Learning Analytics Dashboards*

The Learning Analytics Dashboards transform the data gathered from education into valuable visual insights, helping with decision making. Interactive dashboards assist educators to analyse student engagement, detect learning trends, and plan using visualized data [15], [17]. Nevertheless, existing systems concentrate solely on the visualization of data and do not include the attendance collection, attendance evaluation, and academic risk prediction.

D. *Research Gap*

Studies conducted in the past have achieved the goals of QR-based attendance system, prediction learning analytics, and dashboards for learning analytics, but there is very little integration of all the above elements. In this study, a system will be developed by integrating all of those into a unified platform to provide effective monitoring of academics.

III. SYSTEM DESIGN

The system uses a three-layer architecture with Laravel PHP, MySQL, and Bootstrap 5 with JavaScript. The Academic Intelligence Layer involves the Attendance Evaluation Engine and Risk Prediction Engine for analysis of academic information. The flowchart of the entire system is presented in Figure 1.

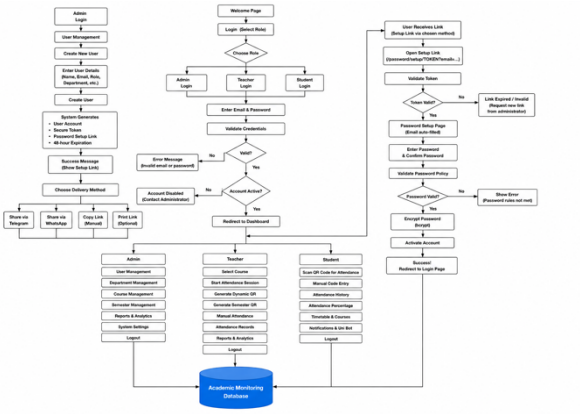


Figure 1. System Flowchart of Proposed System

A. QR Attendance System

The system supports two QR-based attendance methods:

1) Dynamic (Session) QR Code: For each session, a unique QR code is generated. The QR code is timestamped and hashed using SHA-256.

2) Static (Semester) QR Code: In each course, a QR code has been generated. It is valid for the entire duration of the semester.

QR Payload Format:
https://IP:8443/student/scan/process?token=SHA256_HASH&session=SESSION_ID

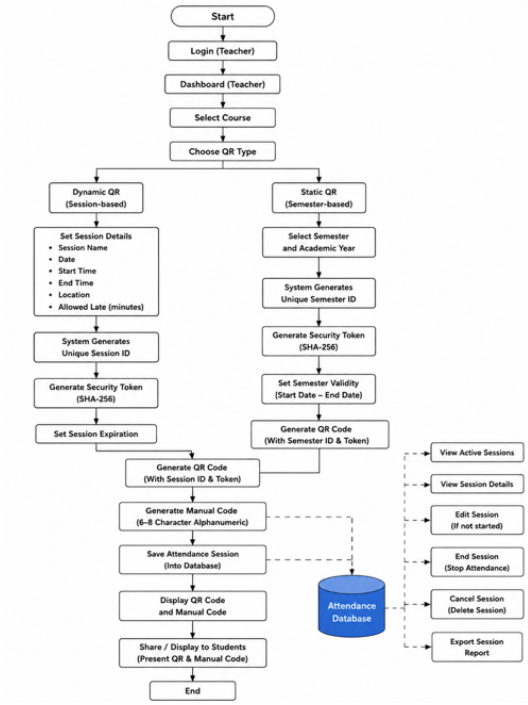


Figure 2. QR Generation Workflow

3) Student Scanning Workflow: The students can scan the QR code scanning via the html5-qr code module. The system will validate the attendance session, QR code, and duplicates. Those students who scan the QR codes during the attendance session will be marked as "Present", and those who scanned after that as "Late", as seen in Figure 3.

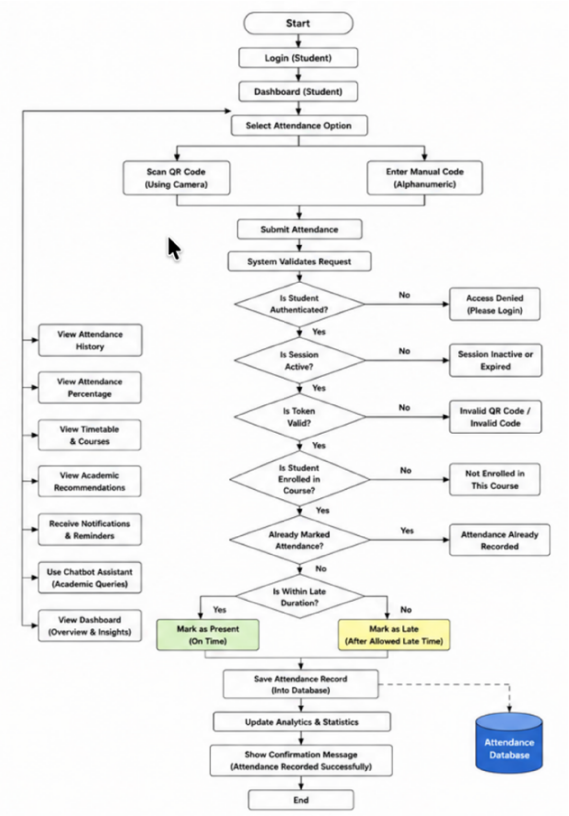


Figure 3. QR Scanning Workflow

B. Attendance Percentage Evaluation

The system meets the attendance eligibility criteria set by the Myanmar Ministry of Education, ensuring standardization and consistency in evaluating students. Attendance percentage follows the following formula:

Attendance Percentage = (Attended Sessions / Total Conducted Sessions) × 100. (1)

C. Eligibility Status Thresholds

The system is used to evaluate the attendance rate whether the student is eligible for examinations according to Myanmar university policies. Those with attendance of ⩾75% are considered eligible; 60-74% attendance is a warning; while <60% is considered at risk.

D. Risk Prediction Framework

The system uses a rules-based risk analysis model to determine the students who require academic intervention because of their attendance. The students are divided into low-risk (⩾75%),medium-risk (60-74%) and high-risk (<60%) depending upon their attendance percentage.

E. Database Schema

The system uses a normalized relational database with 20+ tables. Figure 4 shows the Entity-Relationship (ER) diagram of the core tables.

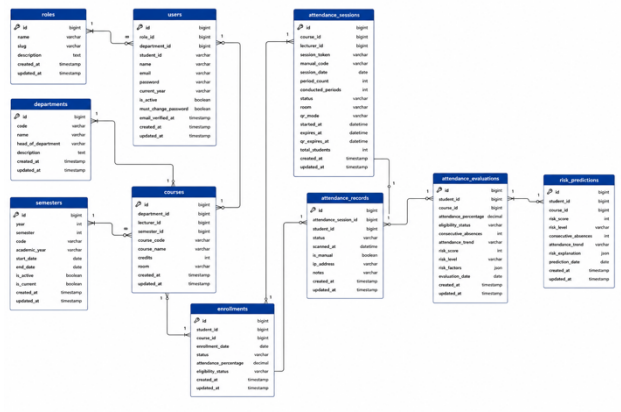


Figure 4. ER Diagram of the Proposed System

F. Security Implementation

This system provides various methods to ensure security of data and unauthorized access. The Laravel bcrypt will encrypt the user's password, while the CSRF tokens and Laravel Middleware will secure the form and role-based access. In addition, time-limited QR codes are validated at the server side before attendance submission.

IV. Role-Based Features

The proposed system serves three distinct stakeholder groups: admins, teachers, and students. Each role has access to role-specific dashboards and functionalities that address their unique requirements in academic ecosystem.

A. Admin Features

The admin manages the overall academic structure, user accounts, and system operations. The dashboard provides a university-wide overview with real-time statistics, department rankings, risk distribution charts, and automated intelligence insights as shown in Figure 5.

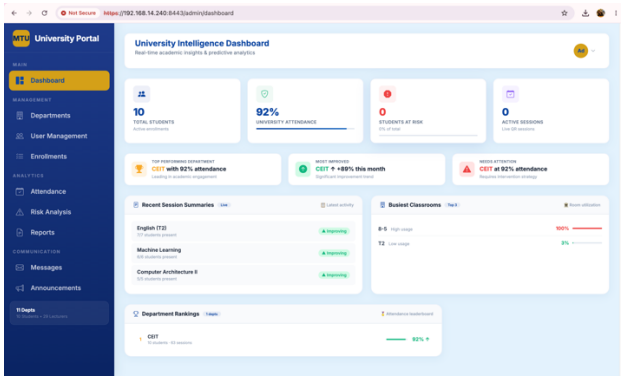


Figure 5. Admin Dashboard

1) User Management: User account management happens through an approval-based system. Accounts will be in Pending status until the admin gets the notification and activates them via email. After activation, the user can log into the system using the password set up during registration.

2) Academic Management: The Academic Management module is used for managing the academic structure of the university through management of departments, and courses. It helps the admins to maintain the academic records.

3) Enrolment Management: Enrolment Management module helps in managing course enrolment of students and approval/rejection of requests. It also shows enrolment status (Pending, Approved, Rejected) as well as enrolment statistics.

4) Attendance Analytics: Attendance Analytics module offers dashboard and reports for tracking university attendance. This includes overall attendance, department ranking, student risk profile, and examination eligibility.

5) Risk Analysis: Risk Analysis module helps in identifying potential academic risks among students based on their attendance percentage. This includes distribution of risk level, department-wise analysis, and at-risk student registration as illustrated in Figure 6.

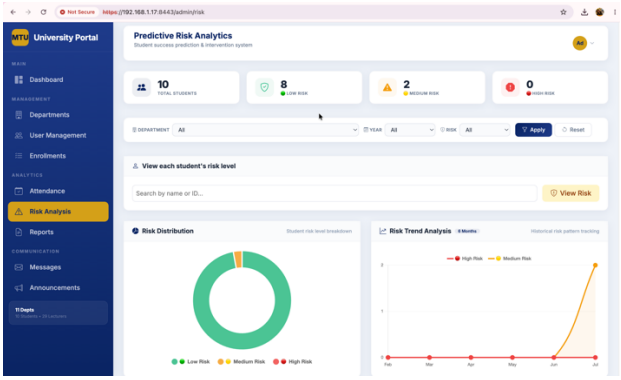


Figure 6. Risk Analysis

6) Reports and Exports: The Reports & Export section allows the admin to create complete reports based on information related to students, attendance, department, and risk analysis. Reports can be exported in CSV and MS Excel format.

B. Teacher Features

The teacher manages classroom sessions, monitors student attendance, and provides academic support. The dashboard provides real-time attendance statistics, active sessions, and at-risk student lists as shown in Figure 7.

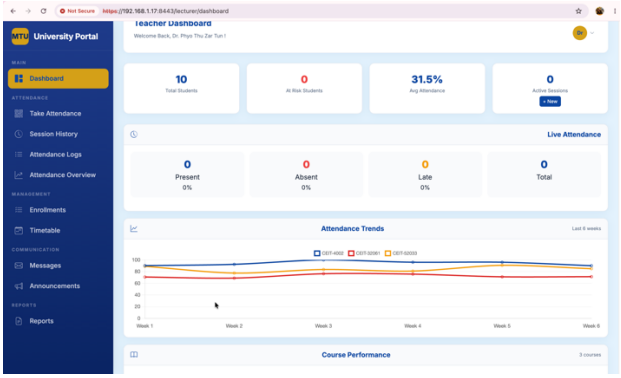


Figure 7. Teacher Dashboard

1) QR Attendance Management: This function permits the use of Dynamic or Static QR codes to register attendance by teachers. Dynamic QR are created for each class, valid for a certain period, whereas a 6-digit alpha-numeric code may be used by those who are not able to use QR codes.

2) Live Attendance Monitoring: Live Attendance Monitoring module gives real-time updates on the attendance status during class. The module shows students present, late, absent, and also makes provision for attendance and manually registering attendance if QR scanning is not possible.

3) Attendance Overview: Attendance Overview module helps teachers keep a check on the attendance in assigned courses. It allows filtering of records either weekly, monthly, semester-wise or date-wise, and also gives details of attendance percentage, eligibility for exams, and students risk level.

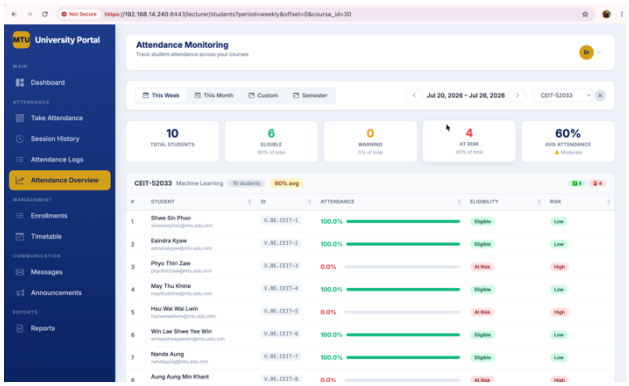


Figure 8. Attendance Overview

4) Risk Monitoring: Risk Monitoring feature helps teachers identify students who may require early academic support. It displays each student's risk level, enabling teachers to provide timely guidance and motivation.

5) Timetable Management: Timetable management module enables teachers with their weekly class schedules, including time and location. Teachers can manage lecture sessions and export timetable data in CSV and PDF formats.

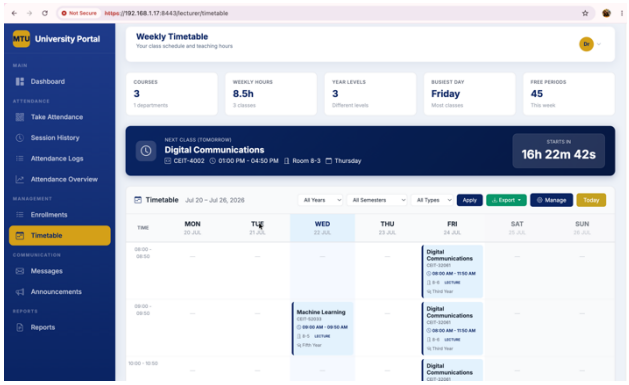


Figure 9. Timetable Management

6) Course Performance and Reports: Course Performance and Reports module helps the teacher to receive information about the students' engagement and attendance. Reports on attendance and at-risk students can be downloaded in PDF, Excel, and CSV formats.

C. Student Features

Student Dashboard provides personalized academic information and progress tracking. It displays academic performance and provides related features, including Academic Assistant Chatbot, as shown in Figure 10.

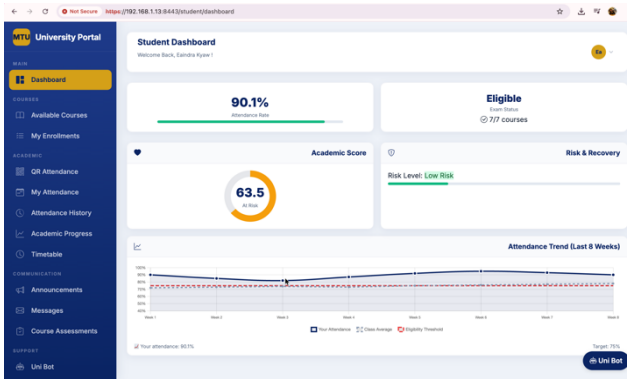


Figure 10. Student Dashboard

1) QR Attendance: QR Attendance feature allows students to record attendance using their mobile phones. They need to scan QR code using HTML5 QR code library, while the system will authenticate the QR code, session validity, and expiration time. In case of failure to scan the QR code, the user can use the attendance code.

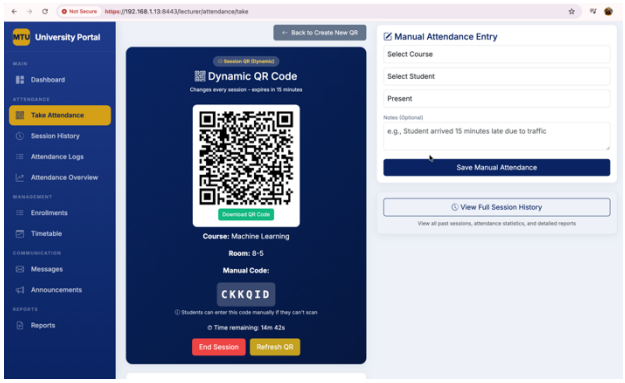


Figure 11. Teacher QR Attendance Generation Interface

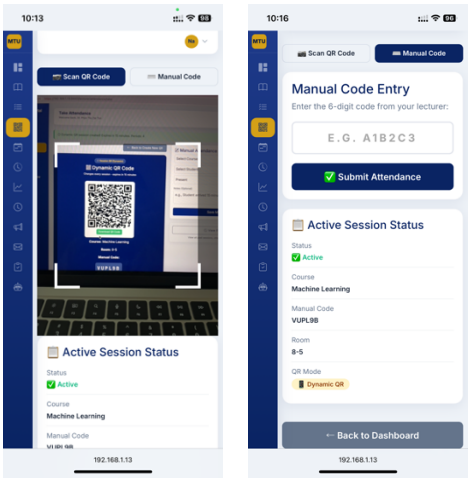


Figure 12. QR Scanning and Manual Attendance Code Entry Interface

2) Academic Monitoring: Academic Monitoring feature provides students with a detailed view of their academic attendance and engagement throughout the semester. It displays overall and course-wise attendance, and risk level.

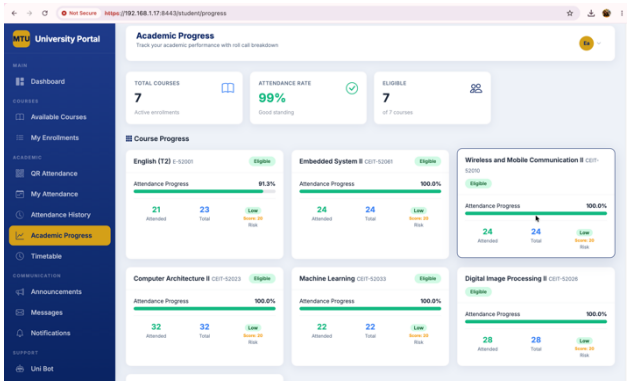


Figure 13. Student Academic Progress

3) Enrolments: Enrolments module allows students to choose from available courses according to their department, session, and semester. They can also make enrolment requests through this module and track their enrolment status.

4) Timetable: Through the Timetable module, the weekly schedule of the courses that a student is taking is provided to him/her, along with information about their classrooms.

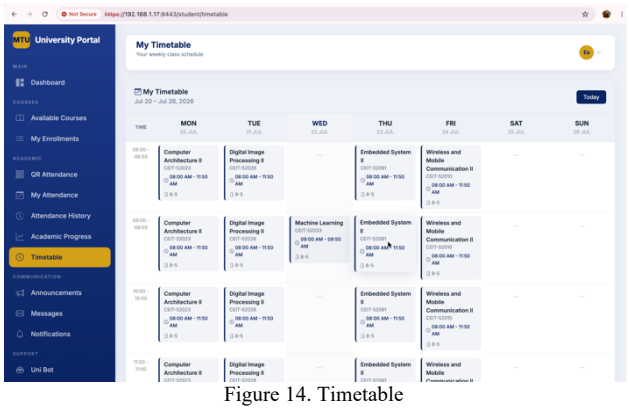


Figure 14. Timetable

5) Risk: The Risk module offers students an insight into their risk level, expected attendance, and other associated causes. This enables the students to comprehend their academic risk and increase their attendance before their performance is adversely affected.

6) Academic Assistant Chatbot: The Academic Assistant Chatbot offers students quick access to their attendance percentage, exam eligibility, risk state, and improvement recommendations.

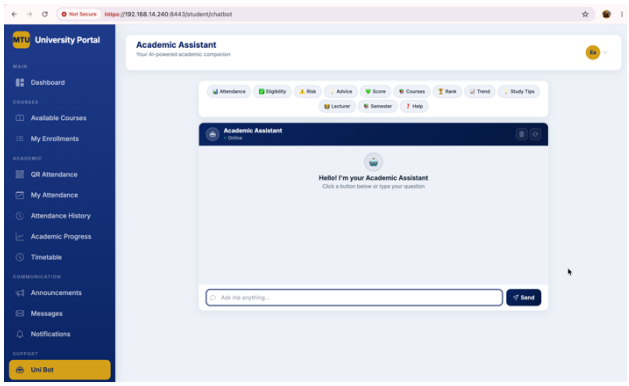


Figure 15. Academic Assistant Chatbot

V. RESULTS AND EVALUATION

A. Test Environment

The system was tested on a university campus with real student and course data from the 2025-2026 academic year at Mandalay Technological University.

Test Data:

1. 10 CEIT students (Fifth Year)
2. 7 CEIT courses
3. 50+ attendance sessions
4. 500+ attendance records

B. Function Testing Results

This testing was performed to confirm that all modules fulfilled their functional requirements such as QR creation and scanning, attendance calculation, risk prediction, and dashboard visualization. All modules had a success rate of 100%, except for risk prediction module that had a success rate of 92% as shown in Table I.

TABLE I
FUNCTION TESTING RESULTS

Feature	Test Cases	Passed	Success Rate
Authentication (Login/Registration)	15	15	100%
QR Generation & Scanning	20	20	100%
Manual Code Fallback	10	10	100%
Risk Prediction	25	23	92%
Dashboard Data Accuracy	15	15	100%

C. Performance Testing Results

Performance testing indicated an average time for capturing attendance of 5 seconds, for QR code scanning and verification of 0.8 seconds, and for the dashboard to load of 1.2 seconds. The number of student scan requests simultaneously supported by the application was 150, surpassing the required 100, with 98% duplicate detection rate and 99.8% system availability rate.

D. Success Metrics

There was an achievement of 100% accuracy in attendance eligibility detection and 92% in risk detection and attendance irregularity detection respectively.

E. Usability Testing Results

Usability testing was done for iOS Safari, Android Chrome, Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge. It was successful in all the key functionalities and the main functionalities are available with three clicks from the dashboard.

F. Sample Test Results

The system was tested using 10 students from the Machine Learning class (CEIT-52033). The QR scan achieved 100% accuracy under 1 second while manual entry achieved 100% accuracy in 5 seconds with no duplications. There were four students with 0% attendance, who were classified as high risk. Of the total students, 60% were considered to be low risk and 40% were considered high risk.

VI. Conclusions

The proposed Intelligent University Portal integrates QR attendance, attendance percentage determination, predictive analytics, and academic monitoring into one unifined platform. QR attendance component includes dynamic session QR codes, static QR codes, and manual attendance codes that allow for the secure and time-bound registration of attendance, as well as preventing duplicate submission of the attendance. Attendance percentages of students are determined by the system, which allows determining their status, and identifying students who need academic intervention. Furthermore, role-based dashboards allow admins, teachers, and students to perform operations related to their roles, timetable management, and attendance reporting facilitate academic administration and monitoring. Academic Assistant Chatbot component allows providing academic assistance to students through the portal. Functional, performance, and usability testing of implemented components demonstrated the effectiveness of attendance management and timely academic monitoring performed by the system. Overall, the results demonstrate that the proposed system can provide secure attendance management, real-time academic monitoring, and early identification of students requiring intervention within a single intelligent university platform.

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References

[1] J. Hendrawan, I. D. Perwitasari, and F. Maulana, "QR Code-Based Attendance Systems in Education: A Systematic Literature Review on Data Accuracy and Sustainable School Management," *Proc. Int. Conf. Comput. Sci., Eng., Social Sci., Multi-Discip. Stud.*, vol. 1, pp. 80–87, 2025, doi: 10.64803/cessmuds.v1.14.

[2] M. Aldabagh, "A Review of Students Attendance Management Systems," *Eurasian Journal of Science and Engineering*, vol. 10, no. 3, pp. 136–145, 2024, doi: 10.23918/eajse.v10i3p13.

[3] R. Yunus and B. Bijanto, "Web-Based Lecture Attendance System with Dynamic QR Code Expired Per Session Using Prototype Method," *Journal of Information System*, vol. 11, no. 1, pp. 99–108, May 2026, doi: 10.33633/joins.v11i1.16137.

[4] A. Elango, N. R. Kuraku, P. Sivakumar, and C. R. Kavitha, "Secure and Tamper-Proof Exam Center Attendance System Using NFC and Encrypted QR Codes," in *Proc. 9th Int. Conf. Electron., Commun. Aerosp. Technol. (ICECA)*, Coimbatore, India, 2025, pp. 439–444, doi: 10.1109/ICECA66444.2025.11382777.

[5] O. Ovtšarenko, "Using AI to Forecast Student Dropout Risk in Technical Education Using a Learning Analytics Approach," *Scientific Reports*, vol. 16, Art. no. 14616, 2026, doi: 10.1038/s41598-026-44919-1.

[6] R. A. M. Rodríguez, J. Valencia, O. D. M. Victoria, and A. A. S. Serrano, "Hybrid Machine Learning Pipeline for Predicting and Segmenting Academic Performance at a Mexican Public University," *IEEE Revista Iberoamericana de Tecnologías del Aprendizaje*, 2026, doi: 10.1109/RITA.2026.3684879.

[7] B. Pei, Y. Cheng, A. Ambrose, E. Dziadula, W. Xing, and J. Lu, "LearningViz: A Dashboard for Visualizing, Analyzing and Closing Learning Performance Gaps—A Case Study Approach," *Smart Learning Environments*, vol. 11, Art. no. 56, 2024, doi: 10.1186/s40561-024-00346-1.

[8] T. H. M. de Oliveira, M. de C. Neto, M. Q. Simões, and E. G. Costa, "Learning Analytics for Academic Empowerment: Transforming a Siloed Institution into a Real-Time, Data-Driven Higher Education Ecosystem," in *Proc. 17th Int. Conf. Educ. New Learn. Technol. (EDULEARN25)*, Palma, Spain, 2025, pp. 10139–10150, doi: 10.21125/edulearn.2025.2635.

[9] S. V. S. H. Reddy et al., "Design of QR Based Smart Student Attendance System," in *Proc. 2023 IEEE 2nd Int. Conf. AI Cybersecurity (ICAIC)*, 2023, doi: 10.1109/ICAIC57335.2023.10044175..

[10] R. Priyadharshini and M. Rathi, "Survey on Smart Attendance Tracking System," *International Journal of Multidisciplinary Research in Science, Engineering and Technology*, vol. 8, no. 4, 2025.

[11] H. S. Brdesee, W. Alsaggaf, N. Aljohani, and S.-U. Hassan, "Predictive Model Using a Machine Learning Approach for Enhancing the Retention Rate of Students At-Risk," *International Journal on Semantic Web and Information Systems*, vol. 18, no. 1, pp. 1–21, 2022, doi: 10.4018/IJSWIS.299859.

[12] L. Chen, X. Geng, M. Lu, A. Shimada, and M. Yamada, "How Students Use Learning Analytics Dashboards in Higher Education: A Learning Performance Perspective," *SAGE Open*, vol. 13, no. 3, pp. 1–18, 2023, doi: 10.1177/21582440231192151.

[13] G. Singh, V. Rastogi, A. Singh, M. K. Nath, and S. Bhadula, "Efficiency and Reliability in Student Attendance: A Trifecta of QR Codes, Face Recognition, and Geolocation," in *Proc. 2024 IEEE Students Conf. Eng. Syst. (SCES)*, vol. 2, pp. 1–5, 2024, doi: 10.1109/SCES61914.2024.10652389.

[14] S. Manickam, M. Al Ismaili, R. C. Sivamani, and S. Dileesh, "An Integrated System for Just-in-Time Formative Assessment and Attendance Monitoring," *Discover Education*, vol. 4, Art. no. 483, 2025, doi: 10.1007/s44217-025-00960-2.

[15] T. Susnjak, "Beyond Predictive Learning Analytics Modelling and onto Explainable Artificial Intelligence with Prescriptive Analytics and ChatGPT," *International Journal of Artificial Intelligence in Education*, vol. 34, pp. 452–482, 2024, doi: 10.1007/s40593-023-00336-3.

[16] D. Getty, M. Turner, M. Nickols, and H. Bisgin, "Leveraging Learning Management Systems for Academic Risk Prediction Using Machine Learning: A Computer Science and Engineering Perspective," in *Proc. IEEE Frontiers in Education Conf. (FIE)*, 2025, doi: 10.1109/FIE63693.2025.11328396.

[17] S. Vemulapalli, P. Bhagat, S. Shukla, N. Jhala, D. Jayanthi, and D. Pulugu, "Predicting Student Performance and Academic Success in Higher Education Using a Hybrid XGBoost-LSTM Model," in *Proc. 2025 8th Int. Conf. Comput. Methodologies Commun. (ICCMC)*, 2025, doi: 10.1109/ICCMC65190.2025.11140772.

[18] Y. Tan, W. H. Orman, V. Pavlopoulos, H. Pham, and R. Patnayakuni, "Transforming Student Data Into Timely Interventions: A Dynamic Bayesian Network Decision Support System," *International Journal of Knowledge Management*, vol. 22, no. 1, 2026, doi: 10.4018/IJKM.398541.

[19] S. Abouelnour, A. Al Redhaei, M. A. Al-Betar, and G. Al-Naymat, "Machine Learning in Higher Education: Predicting and Mitigating Student Dropout," in *Proc. 2024 25th Int. Arab Conf. Inf. Technol. (ACIT)*, Zarqa, Jordan, 2024, pp. 1–7, doi: 10.1109/ACIT62805.2024.10877099.